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SEN201 - SOFTWARE ENGINEERING

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)
FOR
INVENTORY AND WAREHOUSE MANAGEMENT SYSTEM

ACADEMIC SESSION: 2025/2026

DATE: 07/05/2026

TABLE OF CONTENTS

Title Page	i
Table of Contents	ii
1. INTRODUCTION	1
1.1 Purpose of the System	1
1.2 Scope of the System	1
1.3 Objectives of the System	2
1.4 Problem Statement	2
1.5 Significance of the Study	3
1.6 Definitions, Acronyms and Abbreviations	3
2. OVERALL DESCRIPTION	4
2.1 Product Perspective	4
2.2 Product Functions	4
2.3 User Classes and Characteristics	5
2.4 Operating Environment	5
2.5 Design and Implementation Constraints	5
2.6 Assumptions and Dependencies	5
3. SYSTEM REQUIREMENTS	6
3.1 Functional Requirements	6
3.2 Non-Functional Requirements	7
4. SYSTEM DESIGN SPECIFICATION	8
4.1 Architectural Design	8
4.2 Database Design	8
4.3 System Modules	8
4.4 User Interface Design	8
5. SOFTWARE ENGINEERING DESIGN CONCEPTS	9
5.1 Incremental Process Model	9
5.2 Control Hierarchy	9
5.3 Structural Partitioning	9
5.4 Coupling and Cohesion	9

6. SYSTEM MODELS AND DIAGRAMS	10
6.1 Use Case Diagram	10
6.2 Entity Relationship Diagram (ERD)	10
6.3 Activity Diagram	10
 7. REFERENCES	 10

1. INTRODUCTION

1.1 Purpose of the System

The purpose of this Software Requirements Specification (SRS) document is to define the requirements for the Inventory and Warehouse Management System. This document describes the functional requirements, non-functional requirements, system features, user roles, and operational requirements needed for the successful development and implementation of the system.

The Inventory and Warehouse Management System is designed to automate inventory and warehouse operations such as product management, stock monitoring, goods receipt processing, goods dispatch processing, low-stock alerts, inventory valuation, and report generation. The system aims to improve inventory accuracy, reduce operational errors, and enhance warehouse management efficiency.

1.2 Scope of the System

The Inventory and Warehouse Management System is a web-based application developed to manage inventory operations and warehouse activities efficiently. The system provides a centralized platform for managing products, monitoring stock levels, recording inventory transactions, and generating inventory reports.

The system supports product and category management, goods receipt and dispatch processing, low-stock monitoring, inventory valuation, and warehouse reporting. The project focuses mainly on inventory and warehouse management operations and does not include advanced enterprise resource planning or e-commerce functionalities.

The system will be implemented using Laravel Framework, MySQL Database, Blade templates, and Tailwind CSS to ensure maintainability, usability, and security.

1.3 Objectives of the System

The main objective of the Inventory and Warehouse Management System is to develop a computerized platform that improves inventory tracking and warehouse management operations.

The specific objectives of the system are:

1. To develop a web-based inventory management system.
2. To implement product and category management using SKU codes.
3. To provide real-time stock monitoring within the warehouse.

4. To implement goods receipt and goods dispatch processing.
5. To generate low-stock alerts and inventory reports.
6. To support inventory valuation and stock movement tracking.
7. To improve inventory accuracy and operational efficiency.
8. To apply software engineering concepts such as the Incremental Process Model, coupling and cohesion, and structural partitioning.

1.4 Problem Statement

Many organizations still manage inventory and warehouse operations using manual methods such as notebooks, spreadsheets, and handwritten records. These methods are inefficient, time-consuming, and prone to human errors.

Manual inventory systems often result in inaccurate stock records, delayed report generation, stock discrepancies, and poor monitoring of goods movement. Warehouse managers may find it difficult to determine the exact quantity of products available at a given time, leading to stock shortages, overstocking, and poor decision-making.

In addition, manual systems do not provide automated low-stock alerts, real-time inventory monitoring, or accurate inventory valuation. These challenges reduce operational efficiency and affect inventory management performance.

Therefore, there is a need for a computerized Inventory and Warehouse Management System that automates inventory tracking, warehouse operations, and reporting processes to improve efficiency, accuracy, accountability, and decision-making.

1.5 Significance of the Study

The Inventory and Warehouse Management System is significant because it improves inventory management operations through automation and centralized data management.

The system improves inventory accuracy by reducing manual errors associated with stock recording and inventory calculations. It also improves operational efficiency by automating inventory transactions, stock monitoring, and report generation.

The system supports better decision-making through inventory reports, low-stock alerts, and inventory valuation features. It also improves accountability and security through role-based access control.

From an academic perspective, the project demonstrates the practical application of software engineering concepts such as the Incremental Process Model, coupling and cohesion, control hierarchy, and software design principles.

1.6 Definitions, Acronyms and Abbreviations

1. Inventory: Goods or products stored within a warehouse for business operations.
2. Warehouse: A storage facility where inventory items are received, stored, and dispatched.
3. SKU: Stock Keeping Unit used for identifying products.
4. Goods Receipt: The process of receiving inventory into the warehouse.
5. Goods Dispatch: The process of removing or distributing inventory from the warehouse.
6. Inventory Valuation: The process of calculating the financial value of inventory.
7. SRS: Software Requirements Specification.
8. MVC: Model-View-Controller software architecture.
9. RBAC: Role-Based Access Control used for system security.
10. Laravel: A PHP framework used for backend development.
11. MySQL: A relational database management system used for storing system data.

2. OVERALL DESCRIPTION

2.1 Product Perspective

The Inventory and Warehouse Management System is a web-based application developed to improve inventory management and warehouse operations. The system provides a centralized platform for managing products, monitoring stock levels, recording inventory transactions, and generating warehouse reports.

The system replaces manual inventory management methods such as notebooks and spreadsheets with an automated solution that improves inventory accuracy, operational efficiency, and decision-making. The system also supports role-based access control for secure warehouse management operations.

2.2 Product Functions

The major functions of the Inventory and Warehouse Management System include:

1. User authentication and role-based access control.
2. Product and category management using SKU codes.
3. Goods receipt processing with automatic stock updates.
4. Goods dispatch processing and stock deduction.
5. Real-time inventory monitoring and stock tracking.
6. Low-stock alert generation and reorder notifications.
7. Inventory valuation and stock movement tracking.
8. Inventory report generation and export functionality.

9. Staff and user account management.

2.3 User Classes and Characteristics

The system supports multiple categories of users with different responsibilities and access permissions.

Super Admin:

The Super Admin manages the overall system, user roles, warehouse settings, and system configurations.

Admin / Warehouse Manager:

The Admin manages products, stock levels, inventory transactions, warehouse reports, and warehouse staff activities.

Salesperson / Warehouse Staff:

Warehouse Staff record goods receipt and dispatch transactions and monitor inventory information assigned to them.

2.4 Operating Environment

The system will operate within a web-based environment and can be accessed through modern web browsers.

Frontend Technologies:

1. HTML
2. CSS
3. Tailwind CSS
4. JavaScript
5. Blade Templates

Backend Technologies:

1. PHP
2. Laravel Framework

Database:

1. MySQL Database
2. Supported Operating Systems:
3. Windows
4. Linux
5. macOS

2.5 Design and Implementation Constraints

The system development and implementation are subject to several constraints:

1. Limited project development time.
2. Limited hardware and software resources.
3. Dependence on web technologies such as Laravel and MySQL.
4. Requirement for internet or local network connectivity.
5. Need for a simple and user-friendly interface for warehouse staff.
6. Security requirements for protecting inventory records and user accounts.

2.6 Assumptions and Dependencies

The system development is based on the following assumptions and dependencies:

1. Users have basic computer knowledge.
2. Warehouse staff will enter accurate inventory data.
3. The server and database systems will remain operational during usage.
4. The system depends on Laravel Framework and MySQL Database.
5. The system requires modern web browsers for proper operation.
6. Stable internet or local network connectivity is available for system access.

3. SYSTEM REQUIREMENTS

3.1 Functional Requirements

The Inventory and Warehouse Management System shall provide the following functionalities:

1. **User Authentication** The system shall allow authorized users to log into the system securely using usernames and passwords.
2. **Product and Category Management** The system shall allow administrators to create, update, delete, and manage products and categories using unique SKU codes.
3. **Goods Receipt Processing** The system shall allow warehouse staff to record incoming inventory transactions and automatically update stock quantities.
4. **Goods Dispatch Processing** The system shall allow warehouse staff to process outgoing inventory transactions and automatically deduct stock quantities.
5. **Inventory Monitoring** The system shall provide real-time monitoring of stock levels and inventory movement within the warehouse.

6. **Low-Stock Alert System** The system shall generate alerts when inventory quantities fall below predefined reorder levels.
7. **Inventory Valuation** The system shall calculate inventory value based on available stock quantity and product cost.
8. **Report Generation** The system shall generate inventory reports such as stock movement reports, low-stock reports, and inventory valuation reports.
9. **Staff and User Management** The system shall allow administrators to manage user accounts and assign system roles and permissions.

3.2 Non-Functional Requirements

The system shall satisfy the following non-functional requirements:

1. **Security** The system shall implement authentication, password encryption, and role-based access control.
2. **Performance** The system shall process inventory transactions efficiently and provide reasonable response times.
3. **Reliability** The system shall maintain accurate inventory records and minimize operational errors.
4. **Usability** The system shall provide a simple, user-friendly, and responsive interface.
5. **Maintainability** The system shall support future updates, maintenance, and scalability.
6. **Availability** The system shall remain accessible during operational periods with minimal downtime.

4. SYSTEM DESIGN SPECIFICATION

4.1 Architectural Design

The Inventory and Warehouse Management System follows the Model-View-Controller (MVC) architectural pattern provided by the Laravel Framework.

The system architecture consists of:

1. Presentation Layer Handles the user interface using Blade templates, Tailwind CSS, and JavaScript.
2. Application Layer Handles business logic, authentication, authorization, and inventory processing using Laravel Framework.
3. Data Layer Handles data storage and retrieval using MySQL Database.

4.2 Database Design

The system uses a relational database structure to manage inventory and warehouse records.

Major database tables include:

1. users
2. products
3. categories
4. Warehouses
5. Stock_levels
6. Goods_receipts
7. Dispatches
8. stock_movements

The database design supports efficient inventory tracking, stock monitoring, and transaction management.

4.3 System Modules

The system consists of the following modules:

1. Authentication Module
2. Product and Category Management Module
3. Goods Receipt Module
4. Goods Dispatch Module
5. Inventory Monitoring Module
6. Low-Stock Alert Module
7. Inventory Valuation Module
8. Reporting Module
9. Staff and User Management Module

4.4 User Interface Design

The system provides a responsive and user-friendly interface developed using Blade templates and Tailwind CSS.

The major interfaces include:

1. Login Page
2. Dashboard Interface
3. Product Management Interface
4. Inventory Monitoring Interface
5. Goods Receipt and Dispatch Interface
6. Report Generation Interface

The interface is designed to support easy navigation and improve operational efficiency for warehouse users.

5. SOFTWARE ENGINEERING DESIGN CONCEPTS

5.1 Incremental Process Model

The system development follows the Incremental Process Model where the application is developed and improved in multiple stages.

Increment 1:

Development of core inventory management features such as authentication, product management, and stock tracking.

Increment 2:

Implementation of goods receipt, goods dispatch, low-stock alerts, and reporting features.

Increment 3:

Implementation of inventory valuation, warehouse monitoring, and advanced inventory analysis functionalities.

5.2 Control Hierarchy

The system uses a hierarchical control structure to organize modules and system operations.

The control hierarchy defines the relationship between major system modules and submodules such as authentication, inventory processing, and reporting modules.

The hierarchy improves system organization, modularity, and maintainability.

5.3 Structural Partitioning

The system applies structural partitioning to separate system functionalities into independent modules.

Horizontal Partitioning:

Separates major functionalities such as inventory management, reporting, and user management.

Vertical Partitioning:

Separates control modules from processing modules to improve maintainability and reduce system complexity.

Structural partitioning improves software organization, modularity, and scalability.

5.4 Coupling and Cohesion

Coupling and cohesion are important software engineering concepts used to measure the quality, maintainability, and modularity of software systems. The Inventory and Warehouse Management System is designed to achieve high cohesion and low coupling between system modules.

Functional Cohesion:

The system modules are designed with functional cohesion, where each module performs a specific and related task. For example, the Inventory Monitoring Module is responsible only for monitoring stock levels and inventory movement, while the Report Generation Module handles inventory report processing. This improves system organization and maintainability.

Data Coupling:

The system modules communicate through structured data such as product IDs, stock quantities, transaction records, and user information. Modules exchange only the required data needed for processing operations. This form of communication represents data coupling and reduces unnecessary dependencies between modules.

Avoidance of Common Coupling:

The system avoids common coupling by preventing modules from sharing global variables directly. Each module accesses information through controlled database operations and defined system processes. This improves security, maintainability, and system reliability.

The application of proper coupling and cohesion principles improves software quality by making the system modular, easier to maintain, scalable, and easier to test during implementation and future upgrades.

6. SYSTEM MODELS AND DIAGRAMS

6.1 Use Case Diagram

The Use Case Diagram describes the interaction between users and the Inventory and Warehouse Management System. The major actors in the system are Super Admin, Admin / Warehouse Manager, and Warehouse Staff.

The major use cases include:

1. User Login and Authentication
2. Product and Category Management
3. Goods Receipt Processing
4. Goods Dispatch Processing
5. Inventory Monitoring
6. Low-Stock Alert Monitoring
7. Inventory Valuation
8. Report Generation
9. Staff and User Management

The Use Case Diagram illustrates how users interact with the system functionalities based on their assigned roles and permissions.

6.2 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) describes the relationship between the major database entities used within the Inventory and Warehouse Management System.

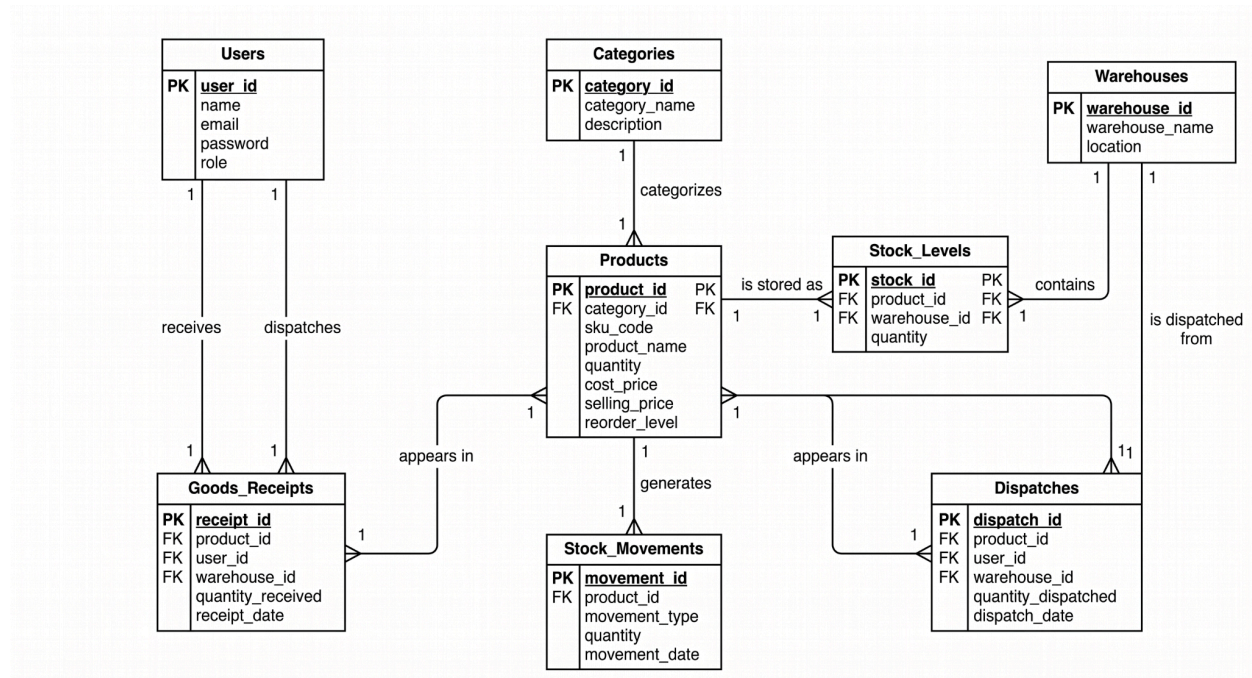


Figure 6.2: Entity Relationship Diagram (ERD)

The major entities include:

1. Users
2. Products
3. Categories
4. Warehouses
5. Stock Levels
6. Goods Receipts
7. Dispatches
8. Stock Movements

Relationships between these entities support inventory tracking, warehouse operations, stock monitoring, and transaction management within the system.

The ERD helps in organizing database structures and maintaining data consistency throughout the application.

6.3 Activity Diagram

The Activity Diagram describes the workflow and operational processes performed within the Inventory and Warehouse Management System.

The major activities include:

1. User Authentication Process
2. Product Registration Process
3. Goods Receipt Workflow
4. Goods Dispatch Workflow
5. Inventory Monitoring Process
6. Low-Stock Alert Generation
7. Report Generation Workflow

The Activity Diagram illustrates the sequence of activities and system operations performed during inventory and warehouse management processes.

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